

POINTS OF INTEREST!!

1. Dress code
2. Seminar topic
3. Understanding of subject matter
4. Confidence
5. Clear presentation
6. Abstract
7. Reference /citation in body of text
8. Conclusion
9. Coherence of write up
10. Response to technical questions
11. General document formatting
12. Literature review (if a research work)
13. Result discussion (if a research work)
14. Overall presentation of report

IMPORTANT INSTRUCTIONS

- Title page-A page, Centre alignment
- Abstract (single line spacing, italics)-A page, no paragraphing
- Reference in American Psychological Association (APA) format-separate page
- Font size: 14
- Line spacing: 1.5
- Font style, Times New Romans
- 3 copies for defense, 1 corrected copy with soft copy in CD after defense for assessment
- Submission date-4 days after defense i.e. one corrected hard copy and soft copy in a CD.
- Length of document; Not less than 10 pages & not more than 12 pages all inclusive(i.e. the title page, abstract, body of text and references)
- Body of text: justify alignment
- Defense date: **will be announced** (***Subject to change***)

SAMPLE TOPICS

1. Tele conferencing and telecommuting
2. The need for campus wifi-mobile application programs
3. Application of online shopping for educational books
4. Application of health care expert system in decision making
5. The use of multimedia tour in higher institution of learning
6. Perceived effect of web-based journal on teaching and learning process
7. Application of e-commerce in banking industry
8. Application of computer aided language system(CAL) in primary schools
9. Applicability of web-based IT training and self-certification
10. Application of unified modeling language in object oriented programming.
11. Application of Case-Based Reasoning System in Treatment Decision of Diseases
12. Advantages of ICT in Research
13. The impact of ICT in Government and Private Agencies
14. Pervasive Cyber criminality and need for alertness
15. ICT development in Nigeria: a way forward
16. Software Method: Panacea For Combating Internet Fraud In Nigeria
17. Multi-agent software development: Advantages and disadvantages
18. Firewalls and Network security
19. Biometrics

20. Object Oriented Programming
21. Comparison of some software life cycle models
22. Telemetric
23. Neural Network
24. E-Government
25. E-governance
26. DNA Computing
27. Human-Computer-Interaction
28. Secure computing platforms
29. Encryption signatures
30. Cryptography and
31. Any other latest topics you think of in computing.

GENERAL FORMAT OF THE REFERENCE LIST

- i. Start the reference list on a new page, with the word References centred at the top of the page
- ii. Second and subsequent lines of each reference should be indented (hanging indent format)
- iii. The reference list should be listed alphabetically by author and then by year
- iv. Book titles and journal titles should be in italics (preferably) or underlined
- v. The date is the year of publication, not printing
- vi. For a book, the edition is only mentioned if it is other than the first
- vii. The place of publication is the town or city, not the country
- viii. Journal titles should be given in full, not abbreviation.
- ix. Do not put a full stop after a website URL
- x. Be consistent in format, layout, type-face and punctuation

CITATIONS IN THE TEXT

Brief citations are inserted within the text wherever you incorporate another's words, facts, or ideas. Each citation contains only enough information, usually just the author's surname and the year of the source, to enable the reader to find the corresponding source in the reference list.

- i. This same point is made by others (Smith & Davies, 2006).
- ii. Smith and Davies (2006) made this same point.
- iii. In 2006, Smith and Davies made this same point.

Note that "and" is replaced by the ampersand sign ("&") when the authors are given in brackets.

To cite a specific part of a source, indicate the page, chapter, figure or table at the appropriate point in the text. Always give page numbers for quotations.

- (Jones, 1998, chap. 3)
- (Walker, 2005, p. 234)

QUOTATIONS IN THE TEXT

Short quotations of less than 40 words should be incorporated into the text, and the quotation enclosed in double quotation marks.

- "This is a quote of less than forty words" (Smith, 2008, p. 43)

Quotations of more than 40 words should be displayed in an indented block of text, without quotation marks.

- Smith (2008) found that:

In the case of quotations of more than forty words you must display the quotation in an indented block of text without quotation marks and quote the author, year and page number in the text, and include a full reference without page number in the reference list (p. 43).

- Smith (2008) found that:

In the case of quotations of more than forty words you must display the quotation in an indented block of text without quotation marks and quote the author, year and page number in the text and include a full reference without page number in the reference list.

If the quotation consists of more than one paragraph you should indent the first line of each paragraph (p. 43).

BOOKS, SINGLE AUTHOR: FORMAT:

Author's Surname, Initials. (Date of publication). Title (Edition, if not the first). Place of publication: Publisher. e.g.

Boddy, D. (2005). *Management: An Introduction* (3rd ed.). Harlow: Financial Times Prentice Hall.

MULTIPLE AUTHORS: FORMAT

1st Author's Surname, Initials, & 2nd Author's Surname, Initials. (Date of publication). Title (Edition, if not the first). Place of publication: Publisher. e.g.

Clarke, S., & Cooper, C. L. (2004). *Managing the risk of Workplace Stress: Health and Safety Hazards*. London: Routledge.

IN THE TEXT

Name all the authors in the first citation. Beginning with the second reference, name only the first author, then add "*et al.*" e.g.

- First citation (Ponton, Gill, Mercer & Smith, 1993)
- Subsequent citations (Ponton *et al.*, 1993)

BOOKS, NO AUTHOR- Some works such as dictionaries and reference books have no author: FORMAT

Title (Edition, if not the first). (Date). Place of Publication: Publisher. e.g.

Merriam-Webster's Collegiate Dictionary (15th ed.). (2003). Springfield, MA: Merriam-Webster.

IN THE TEXT: (Merriam-Webster's Collegiate Dictionary, 2003)

BOOKS, ELECTRONIC: FORMAT

Author's Surname, Initials. (Date of publication). Title. (Edition, if not the first). Place of publication: Publisher. Retrieved day month, year, from website URL e.g.

Saunders, M., Lewis, P., & Thornhill, A. (2007). *Research Methods for Business Students* (4th ed.). Harlow: Pearson. Retrieved 28 October, 2008, from <http://www.mylibrary.com?id=177101>

JOURNAL ARTICLES, PRINTED: FORMAT

Author's Surname, initials. (Year of Journal). Full title of article. Full Title of Journal, Volume Number (Issue Number*), page numbers of article. e.g.

Bulus L.D., Sajoh D.I. and Onyeka N.C. (2013), Social Engineering Related ATM Fraud: A Way Forward, *Journal of Physical Science and Innovation*, Vol.5, No.1, Pp. 106-115.

Tosy, C. (1996). A Meta-analysis of Consumer Choice and Subliminal Advertising. *Psychology and Marketing*, 13, 517-530.

NEWSPAPER ARTICLES, PRINTED: FORMAT

Author's Surname, initials. (Year, month day). Full title of article. Title of Newspaper, page numbers of article. e.g.

Bowers, S. (2008, November 15). Lehman Administrators' Task Will Dwarf Enron, Creditors Told. Guardian, p. 39.

Seminar paper may be within these two categories:

- Researched paper
- Reviewed paper

**SOCIAL ENGINEERING RELATED ATM FRAUD:
A WAY FORWARD**

BY

**JOEL JOHN
ST/CS/ND/23/301**

A SEMINAR

PRESENTED TO

**THE DEPARTMENT OF COMPUTER SCIENCE,
SCHOOL OF SCIENCE AND TECHNOLOGY,
FEDERAL POLYTECHNIC, MUBI
ADAMAWA STATE, NIGERIA**

JULY, 2024

ABSTRACT

It is well-known that criminals have many ways of illegally accessing ATM card PIN to retrieve money in account of legitimate users. In this seminar, we briefly provide an overview of the possible fraudulent activities that may be associated with social engineering and provide security measures to guide against the problem. One Hundred Questionnaires were distributed to respondents; Fig.3 in appendix illustrates the respondents' views in percentage. More so, Table 3 showed the chi-square distribution tests where we reject the null hypothesis and accept the alternative hypothesis. Hence conclude that the exposure of ATM card PIN to second party make users of ATM card vulnerable to social engineering attacks . Recommendations are made to guide users of ATM card both online and offline and never to disclose their ATM card PIN detail to second party.

INTRODUCTION

Throughout history, criminals have often engaged in unauthorized acquisition and use of another person's identity to obtain some advantage they are not entitled to receive. (Report Public Safety, 2010).

Adewuyi (2011) states that Automated Teller machine (ATM) is basically a cash dispenser, it has a unique service of 24/7 facility, "stand alone" or "wall mounted". Fraudsters deceive and manipulate people, exploring human weaknesses through the use of ATM related fraud to obtain personal benefit especially in time like this. They use social engineering techniques as part of Information Technology (IT) fraud.

Janet (2011) explained that reducing fraud to the minimal will help both the Banks and the customers to understand each other in terms of business. This will also build customer confidence and increase the Bank's reputation in comparison to its competitors.

Social engineering websites sometimes referred to as "friend-of a friend" sites and are built upon the concept of traditional social networks where one is connected to new or already known people. But the area of concern is the security implications posed by the sites. Social engineering websites rely on connections and communication so that they encourage one to provide a certain amount of personal information. When deciding how much information to reveal, people may not exercise the same amount of caution as they would when meeting someone in person (Ukpe, *et al.*, 2011).

A study by Kimberly (2007), clarified that Social engineers use influence and persuasion to deceive people for the purpose of obtaining information to perform some action. He further stated that a social engineer commonly uses the telephone or Internet to trick people into revealing sensitive information. Basically two common types of attacks exist, these are:

- i. **Human-based:** Human-based social engineering refers to person-to-person interaction to retrieve the desired information. An example is calling the help desk and trying to find out a password.
- ii. **Computer-based:** Computer-based social engineering refers to having computer software that attempts to retrieve the desired information. An example is sending a user an e-mail and asking them to re-enter a password in a web page to confirm, it employs a computer in the attack (Ukpe *et al.*, 2011).

Phishing technique is said to be mostly used by hackers to obtain confidential details of bank password via email and sometimes they collect ATM card PINs. The customer believes in the email looking exactly like his banker's asking for personal details, password, PIN etc., and blindly complies with the request and through that they invade into his account, (Subramanian, 2002).

Sarah (2010) observed that Social engineering can appear in different forms, depending upon where someone sits. Social engineering is generally a hacker's clever manipulation of the natural human tendency to trust. The hacker's goal is to obtain information that will allow him/her to gain unauthorized access to valuable information.

As pointed by Margaret (2006),

Social engineers rely on the fact that people are not aware of the value of the information they possess and are careless about protecting it. Social engineers take advantage of people's natural understanding to choose passwords that are meaningful to them but can be easily guessed especially with ATM card PIN.

As studied by Ukpe *et al.*(2011), "Phishing" is a criminal mechanism employing social engineering to steal customer's personal identity data and financial account details. It is a form of social engineering that deceives customers' into disclosing their personal and financial data, such as passwords, ATM PIN numbers, credit card numbers and bank account numbers. Among other forms of crime, phishing scams are on the increase in Nigeria.

RESEARCH QUESTION

Are people making themselves vulnerable to social engineering attacks?

Hypothesis

H₀: The exposure of ATM card PIN to second party does not make users of ATM card vulnerable to social engineering attacks

H₁: The exposure of ATM card PIN to second party make users of ATM card vulnerable to social engineering attacks

MATERIAL AND METHOD

This study was carried out with the intention of minimizing the problems associated with social engineering through the use of ATM card. One hundred Questionnaires were issued out to respondents in Federal Polytechnic Mubi to gather their views.

INSTRUMENT FOR DATA COLLECTION

The research instrument is the questionnaire, titled “SOCIAL ENGINEERING RELATED ATM FRAUD: A WAY FORWARD” was administered to respondents.

RESULTS AND DISCUSSION

One hundred questions were issued out, ninety nine (99) questions were returned filled; respondents' sex and age groups were specified.

Using Table 1 & 2, the study revealed that 68% of the populations are men, while only 32 are female which showed that men patronised the use of ATM card than women. The study further examined age groups, 15-24, 25-34, 35-44, and 45 & above representing 12%, 46%, 27% and 14% respectively. From this age group, it can be conclude that youth engaged much in the use of ATM card having 46% of the responses.

Table 4 shows t-test distribution. From Table4, we reject the null hypothesis and accept the alternative hypothesis which, state that users of ATM are making themselves vulnerable to social engineering attacks by exposure it to second party.

Conclusively, we can deduce that, almost all individual has and used ATM card. Exposing their ATM card PIN to second party is an exposure to social engineering attacks.

WAY FORWARD

It is important to note that, social engineering attack are on the increase in term of ATM card PIN exposure in Nigeria, but we can manage the situation if and only if every single person would take the responsibility of his/her actions. Among these are:

- i. Avoid over helpful hands:
- ii. Cover your PIN while withdrawing:
- iii. Monitor accounts regularly:
- iv. Always keep confidential data safe.
- v. Know how to change password, how often to change the passwords, who gets access to your information, and destruction of paper documents regarding confidential information.
- vi. Identifying which information is sensitive and evaluating its exposure to security systems.

CONCLUSION

The seminar is an attempt to equip the users of ATM on the needed to know hidden related social engineering fraud in the ATM system. Fraud is preventable; it can be closely monitored in order to reduce the consequences and frequencies of occurrence in the future.

Always proceed with caution : Guard your Personal information both online and offline; never disclosed your ATM card PIN to anyone, above all to strangers who e-mail you and claim to be from your bank.

ACKNOWLEDGEMENT

All respondents of the questionnaires are highly acknowledged and authors of different journals, books and articles used are also recognized.

REFERENCES

- Adewuyi, I. D. (2011). Electronic Banking in Nigeria: Challenges of the Regulatory Authorities and the way Forward, *International Journal of Economic Development Research and Investment*, 2 (1); 149-156.
- A Report to the Attorney General of the United States and the Minister of Public Safety of Canada (November, 2010); Identity-Related Crime: A Threat Assessment, Access May 10, 2013 from <http://www.justice.gov/criminal/fraud/documents/reports/2010/11-01-10mass-market-fraud.pdf>
- Azeez N. A. & Barry I. (2010). Cyber Security: Challenges and the Way Forward, *GESJ: Computer Science and Telecommunications*. 6(29), Accessed May 2nd 2013 from <http://www.gesj.internet-academy.org>
- Ikem C. O. (2003). *Research Manual, Guide for Research in Applied Science, Education Technology, Medicine, Engineering and Business Studies*, Paraclete Publishers, Yola, Nigeria
- Joan G. (2010, Feb 25). *ATM Skimming: How to Recognize Card Fraud*, CXO Media Inc. a subsidiary of IDG Enterprise , Access May 3 2013 from <http://www.csoonline.com/article/555863/atm-skimming-how-to-recognize-card-fraud>
- Kimberly G. (2007): *Official Certified Ethical Hacker Review Guide*, Wiley Publishing, Inc., Canada ISBN-13: 978-0-7821-4437-6. Pg. 28-37
- Oxford advanced learner's dictionary (8th Edition) 2010) , International Student Edition, Oxford University Press.
- Ukpe K., Salami A., Odulaja, G.O., Mahmood, Z.A, Omotoso R. & Mughele, E. S.(2011 June 13). Social Engineering and Workplace Productivity - Balancing the Odds, *African Journal Computer & Information Communication Technology* ,4(1); 51-54

APPENDIX

Table 3: Common related exposure of The ATM card

No./Responses	Yes	No	Total
1	91	2	93
2	35	56	91
3	42	49	91
4	42	49	91
5	44	47	91
6	19	72	91
7	19	72	91
Total	292	347	639

$$C_{ij} = \frac{(R \times C)}{G}$$

Table4: Chi-square computational table

O	E	O-E	(O-E) ²	(O-E) ² /E
91	42.5	48.5	2352.25	55.34706
35	41.58	-6.58	43.2964	1.041279
42	41.58	0.42	0.1764	0.004242
42	41.58	0.42	0.1764	0.004242
44	41.58	2.42	5.8564	0.140847
19	41.58	-22.58	509.8564	12.26206
19	41.58	-22.58	509.8564	12.26206
2	50.5	-48.5	2352.25	46.57921
56	49.42	6.58	43.2964	0.876091
49	49.42	-0.42	0.1764	0.003569
49	49.42	-0.42	0.1764	0.003569
47	49.42	-2.42	5.8564	0.118503
72	49.42	22.58	509.8564	10.3168
72	49.42	22.58	509.8564	10.3168
				149.2763

X^2 = chi- square value, O = observed value, E = expected value

$$X_{tab}^2 = (R - 1)(C - 1),$$

$$df = (7-1)(2-1) = 3,$$

$$6, 5\% = 12.592$$

$$X_{cal}^2 = \sum_{i=1}^n \frac{(O_{ij} - E_{ij})^2}{E_{ij}}, \quad X_{cal}^2 = 149.28$$

Decision Rule: $X_{cal}^2 149.28 > X_{tab}^2$ that is, $H_0 > H_1$ we reject the H_0 and accept H_1

We reject the null hypothesis and accept the alternative hypothesis. Hence conclude that people are not making themselves vulnerable to social engineering attacks.